

VBF-T1LC-DSH-01

Technical Datasheet

Case: t1_lc | Date: 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Technical Datasheet — VBF-T1LC-DSH-01

Next-generation pure-electric sedan cell (virtual design). Values mechanically taken from parameter set / simulation outputs.

Field	Value	Source
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Rated capacity	5.0 Ah (nominal); 5.03 Ah (simulation-verified)	parameter set / cell/r3_V3_1c_dfn.json
Nominal voltage (plateau)	3.82 V	cell/r3_V3_energy.json:midpoint_voltage_v
Voltage window	2.5 – 4.2 V	parameter set
Rated energy	17.95 Wh	cell/r3_V3_energy.json:energy_wh
Energy density (gravimetric)	413.02 Wh/kg	cell/r3_V3_energy.json:energy_density_wh_kg
Energy density (volumetric, contract)	870.31 Wh/L	cell/r3_V3_energy.json:energy_density_wh_l
Maximum continuous discharge	1C (5.03 A)	1C discharge simulation
Fast-charge capability	4C charge: T_max 55.4 deg C, no lithium plating	cell/r3_V3_4c_dfn.json
DC resistance	3.38 mΩ	cell/r3_V3_energy.json:dcr_ohm
Power density (theoretical peak)	28.33 kW/kg	cell/r3_V3_energy.json:power_density_w_kg
Operating temperature range	discharge 25 deg C; fast charge 45 deg C ambient (simulated conditions)	simulation protocol
Cycle life	Not simulated (aging model not run for this case)	honest N/A
Safety determination	no plating at 4C; T_max <= 60 deg C; overcharge 4.7 V no thermal runaway	cell/r3_V3_4c_dfn.json / cell/r3_V3_tr.json
Dimensions	electrode sheet 65 mm x 1580 mm; total layer thickness 200.8 um; shell not provided	parameter set / energy output
Mass	43.45 g (contract caliber, electrolyte excluded)	cell/r3_V3_energy.json:mass_kg

Notes:

- Gravimetric energy density is contract caliber: discharge energy / layer mass (electrode active + current collectors + separator), electrolyte and casing excluded.
- Cycle life and physical-abuse items require physical experiment / aging simulation not run here; reported honestly as not provided.